KeyBench: A Resource for Extracting Keywords from Scientific Papers

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Abstract

This paper describes a resource to encourage the community to build tools to extract keywords. Many papers on ArXiv list keywords on the first page, typically near the abstract. The proposed resource scrapes these keywords, and uses them as the gold standard for a prediction task. An evaluation compares three baselines for predicting the gold standard: (1) VitaLity, (2) topics from Semantic Scholar and (3) a chatbot. The resource is intended to challenge the community to beat these baselines. To make it easier for the community to build such systems, the resource includes a number of fields for 117,877 papers from Semantic Scholar: (a) title, (b) abstract, (c) authors, (d) citing sentences, and (e) ids to make it easier to join with data in resources such as ArXiv, Semantic Scholar, ACL Anthology, PubMed, etc. We hope the community will show that citing sentences are useful because of the wisdom of the crowd. Good keywords are likely to be used by many authors.

CCS Concepts

• **Information systems → Document topic models**; **Content analysis and feature selection**.

Keywords

KeyBench, Keywords, Semantic Scholar, VitaLITy, Resource Paper

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1 Introduction

Keywords play an important role in the scientific literature. There is a considerable body of work on keyword extraction [3, 5, 6, 9, 16, 17, 22, 24, 26]. It is worthwhile revisiting keyword extraction since much of the literature predates recent advances with LLMs.

Many venues encourage authors to select a few representative keywords.[1] KeyBench follows the standard practice in the keyword extraction literature of using author-supplied keywords as the gold standard. SemEval-10[2] [10] combined these keywords from authors with keywords from readers (undergraduate CS students). [9] uses keywords from professional indexers as the gold standard.

The community has made a considerable investment in keywords because it is widely believed that keywords are useful. But what for? Some possible use cases of keywords (and topic clustering) are:

(1) Help search engines.
(2) Help readers quickly figure out how a paper fits into the larger literature.
(3) Help the community identify important topics and standardize the terminology.

As suggested above, the proposed resource, KeyBench, provides author-supplied keywords as a gold standard for a keyword extraction task. KeyBench also contains citing sentences. We believe there is a huge opportunity to take advantage of citing sentences, though the literature on citing sentences [4, 13, 20] is not as large as the literature on keyword extraction. Following [4], we challenge the community to find ways to use citing sentences and the wisdom of the crowd to enhance keyword extraction. That said, we are concerned that citing sentences may not be useful for keyword extraction because authors may not agree with the crowd on terminology, or even, what is important, as will be discussed below.

Moreover, authors and the crowd may be working cross purposes from the perspective of the given-new distinction [19] in pragmatics. Terminology provides a convenient short-hand to refer to background material (given), unlike author-supplied keywords which are often used to introduce new material. Thus, it might be the case that citing sentences can be useful, though for purposes other than enhancing (author-supplied) keyword extraction.

In addition, the literature is a moving target. Author-supplied keywords are determined at publication time, unlike citing sentences, which are continuously evolving, well after publication time. If one wants to know how to refer to a seminal paper in the contemporary literature, citing sentences are more representative of the contemporary literature than the primary source.

A good set of keywords provides a brief summary of key aspects of a paper. Authors provide keywords and phrases that are usually not covered in a pre-existing taxonomy or classification scheme. In academic search, this metadata can be used as an input source for many tasks such as classification, recommendation, indexing, etc. Terminology in citing sentences is similar to keywords but different. The community often uses contemporary terminology and author names (e.g., RSA [21], AWK [1], Turing machines, Hamming codes).

Our main contribution is the resource in footnote 5 and GitHub in footnote 6. The last author created the *S* labels in KeyBench, as well

*Work does not relate to the author's position at Amazon.
[1] https://www.acm.org/publications/computing-classification-system/how-to-use

[2] https://github.com/snkim/AutomaticKeyphraseExtraction

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as the (unpublished) AWS files mentioned in the footnotes in subsection 3.2. Unlike most keyword-extraction resources, KeyBench includes citing sentences to draw attention to differences between terminology in citing sentences and author-supplied keywords.

2 Use Case Scenario

Consider the paper, *Embedding-based Retrieval in Facebook Search* [8]. The keywords provided by the authors, noted as Gold (G), are: embedding, deep learning, search, information retrieval.

If we look at other systems that automatically extract keywords for this paper like VitaLITy (V) and Semantic Scholar (S) we have:

- V: Learning Latent Representations, Machine Learning, Search Engine Architectures And Scalability, Retrieval Models And Ranking, Information Retrieval, Computing Methodologies, Machine Learning Approaches, Information Systems.
- S_m (mentions): embed base retrieval, semantic embedding, social network, facebook search, personalize search, social graph, invert index, web search.
- S_c (cited for): embed base retrieval, facebook search, search retrieval, random negative.

The V and S_m labels come from systems that focus on the abstract:

> Search in social networks such as Facebook poses different challenges than in classical web search: besides the query text, it is important to take into account the searcher's context to provide relevant results. Their social graph is an integral part of this context and is a unique aspect of Facebook search. While embedding-based retrieval (EBR) has been applied in web search engines for years, Facebook search was still mainly based on a Boolean matching model. In this paper, we discuss the techniques for applying EBR to a Facebook Search system. We introduce the unified embedding framework developed to model semantic embeddings for personalized search, and the system to serve embedding-based retrieval in a typical search system based on an inverted index. We discuss various tricks and experiences on end-to-end optimization of the whole system, including ANN parameter tuning and full-stack optimization. Finally, we present our progress on two selected advanced topics about modeling. We evaluated EBR on verticals for Facebook Search with significant metrics gains observed in online A/B experiments. We believe this paper will provide useful insights and experiences to help people on developing embedding-based retrieval systems in search engines.

S_c labels like *random negative* are obtained from intersecting terms from text in citing papers as illustrated in the example below and described in subsection 3.2.

A sample of a few citing sentences for [8]:

- Rand Neg outperforms BM25, DeepCT, and LeToR even by a large margin on some metrics.
- As we expect, random negative sampling methods, namely Rand Neg and In-Batch Neg, well minimize the total pairwise errors but cannot effectively minimize the top-K pairwise errors.
- ADORE with PQ=6 achieves similar MRR@10 with Rand Neg (0.304 vs. 0.301).
- We start with introducing the widely-used random negative sampling method [7, 12, 29].
- For random negative sampling baselines, we present Rand Neg [12] and In-Batch Neg [15, 33].
- Specifically, BM25 Neg model almost underperforms Rand Neg in terms of every metric.

Note that the citing sentences above use the phrase, *Rand Neg*, even though this term does not appear in [8]. The community often comes up with useful short-hand phrases like Rand Neg. Sometimes these terms can be found in the original paper (as uncovered by S_c), but often such terms are introduced later, using conventions such as variants of the authors name(s), e.g., RSA[3] [21], Turing Machine[4] [23]. As will be discussed below, Turing's paper introduced the *Halting Problem*, though he referred to that as *Entscheidungsproblem*. If one wants to know how to refer to a paper in the contemporary literature, the contemporary literature (e.g., citing sentences) is often more helpful than the primary literature (e.g., abstracts).

2.1 More Citing Sentences

We showcase a few citing sentences from two well-known papers. Like the example above, citing sentences ought to be useful for identifying contributions that matter more to contemporary audiences than to the authors of the original paper.

2.1.1 Turing's paper. As mentioned above, citing sentences use terms like Turing Machine and Halting Problem, which are not found in the original paper [23].

- Alan Turing's famous halting problem entails that no general algorithm can determine whether a given arbitrary program will ever finish its execution.
- These abstractions underpin foundational results such as the undecidability of the halting problem (Turing 1936) and remain indispensable to theoretical computer science.
- Turing machines treat computation as a sequence of transitions between states expressed as symbol manipulations on a tape, emphasising how to compute (Turing 1936).
- Turing machines, introduced by Turing in his seminal work (Turing 1936), constitute the earliest and most widely used formal model of computation, and are regarded as the foundation of theoretical computer science.

2.1.2 Penn Treebank. [15] is often cited as a resource for estimating perplexity even though that use has little to do with the authors' interests in parse trees (and Penn).

- The best performance on the MMLU task is achieved with the PTB dataset, while the best results for common sense reasoning tasks are obtained using the Alpaca dataset.
- To illustrate the task-aware effect on compression performance, we evaluate our method using four training datasets: the first 20.48M tokens of the C4 (Raffel et al., 2019) set, the Alpaca (Taori et al., 2023) training dataset, the WikiText-2 (Merity et al., 2016) training set, and the PTB (Marcus et al., 1993) training set.

[3]https://topics-beta.apps.allenai.org/topic/33775014601
[4]https://topics-beta.apps.semanticscholar.org/topic/68584673328

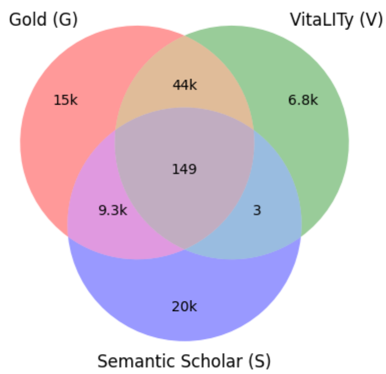

Figure 1: A Venn diagram of 96K of 118K papers in KeyBench (22K papers are not in $|G \cup V \cup S|$). The gold set, G, was constructed to have a large intersection with S. V was chosen by others for other purposes.

- On LSTM, compared to 8-bit uniform quantization PG boosts perplexity per word (PPW) by 1.2% with 2.8× less compute on the Penn Tree Bank (PTB) dataset.
- In recent efforts to construct large 'treebanks' of parsed texts, canonical parsing has been used as a first but small step.

3 Resource Construction and Task Description

The KeyBench resource is posted here;[5] evaluation scripts and more are posted here.[6] KeyBench contains papers with one or more of:

- Keywords: Gold (G), Semantic Scholar (S), VitaLITy (V)
- Title, Abstract, Authors
- Citing sentences
- Ids (for joining with other resources)

Most values are based on Semantic Scholar [12], except for V, G and S labels. V represents VitaLITy[7] [2]; G and S are described below.

The task is to predict the Gold (G) labels from the other fields. The community is encouraged to build systems and compare their predictions with alternatives such as V and S.

Ideally, all systems would be compared on the same documents using a factorial design, but unfortunately, the Venn diagram in Figure 1 makes it clear that some combinations have more coverage than others in KeyBench. We believe it is useful to include V labels in KeyBench even though it was designed for other purposes.[8]

3.1 Gold (G) Labels

Many of the ArXiv papers in S2ORC [14] list keywords on the first page. G labels in KeyBench were obtained by scraping these papers.

[5]KeyBench is available in three places: https://doi.org/10.7910/DVN/CZNUZN, https://doi.org/10.5061/dryad.bk3j9kdsn and https://drive.google.com/file/d/17LEj3Ybwyt_oDq-vjcxmOo065lhC7KVo/view?usp=sharing

[6] https://github.com/kwchurch/KeyBench

[7]https://vitality-vis.github.io/

[8]Since https://vitality-vis.github.io/ does not provide join keys, we inferred Semantic Scholar CorpusIds using a heuristic based on titles.

Encoder	Msec/batch	P	R	F1
SciBERT	57	73.43	75.25	74.33
DistilBERT	39	70.39	73.92	72.11
CS RoBERTa DAPT	**56**	**74.72**	**76.18**	**75.44**
SciDeBERTa CS	85	74.76	76.72	75.73
MiniLM	38	72.94	74.21	73.57

Table 1: Evaluation of NER methods trained on output from bot. The deployed configuration is highlighted in bold.

Fitz[9] offers simple ways to convert pdf files to json. The keywords in KeyBench were extracted from this json using `json2keywords`, a Python program included in the GitHub mentioned in footnote 6.

3.2 Semantic Scholar (S) Labels

KeyBench includes topics from Semantic Scholar based on three publicly available files:

- S_c (cited for) and S_m (mention) labels for ~11.6M papers[10]
- List of ~658K topics[11]
- Descriptions of ~492K topics[12]

The community is strongly encouraged to use these resources for further analysis. However, KeyBench only uses a tiny subset; if we added 11.6M S labels, KeyBench would have been larger (and harder to distribute and process). To make KeyBench more manageable, we excluded papers if there are no labels (other than S).

S labels are based on a pipeline that inputs 12M Computer Science papers from S2ORC and produces topics. For a given paper, S_c labels represent topics that other papers potentially cite it for, and S_m are the subset of topics mentioned in its abstract from the entire set. The pipeline also produces useful information that goes beyond the scope of KeyBench, i.e., LLM generated descriptions of topics.

- Corpus construction: for each paper in the S2ORC CS subset, extract title, abstract, body paragraphs and inline citation spans.
- NER (concept extraction and normalization): An NER approach is used to identify topics. Training data was constructed by sampling abstracts and body paragraphs, and labeling them with `GPT-3.5-turbo`, using the prompt in subsection 3.3. Based on the performance of trained models in Table 1, it was decided to deploy CS RoBERTa DAPT [7]. In addition, there is a special case for acronyms, based on MadDog [18].[13]
- Associate topics with introducing papers as in [11].
- Paper-topic pair ranking/filtering with a lightGBM model.
- Generate topic descriptions from mentions in cited papers.

3.3 Prompt for GPT-3.5-turbo

System.

[9]https://github.com/pymupdf/pymupdf

[10]https://us-west-2.console.aws.amazon.com/s3/buckets/ai2-s2-research-public?prefix=topics_db_04272026/topic_papers_pdp/

[11]https://us-west-2.console.aws.amazon.com/s3/buckets/ai2-s2-research-public?prefix=topics_db_04272026/topics_list/

[12]https://us-west-2.console.aws.amazon.com/s3/buckets/ai2-s2-research-public?prefix=topics_db_04272026/all_descriptions/

[13]For an example of a description, see https://topics-beta.apps.semanticscholar.org/topic/29388308929?corpusId=16486302.

```
You are a researcher with extensive knowledge
about all the scientific fields.
```

User.

```
Output a python list of specific scientific
concepts in the following text as valid lower
case string encased within double quotes. Include
acronyms and abbreviations as well.
{text}
```

Assistant prefix.

```
concept_list=
```

3.4 Evaluation Procedure

Of course, the task of identifying useful short-hand terms is not the same as the keyword prediction task. KeyBench uses keyword prediction because it is easy to construct a credible gold standard, though we are perhaps more interested in identifying short-hand.

The evaluation script in KeyBench scores candidate labels (from V, S as well as other submissions from the community) in terms of precision and recall. As one can see from the labels above, these scores depend on what counts as a match. The evaluation script uses a number of different definitions including (a) exact match, (b) Porter stemmer [25] and (c) edit distance (with a collar).

We believe citing sentences should also be useful, though most systems do not currently take advantage of this opportunity. Preliminary experiments with bots (LLM-based prompts) suggest that citing sentences may not improve prediction of G labels (much, if at all), though we included citing sentences in KeyBench, hoping the community will find a way to take advantage of citing sentences.

3.5 Baseline Results

Figure 2 shows precision/recall for 4 baselines and 3 match types:

1. Exact match [black]: The baseline outputs the same string as a gold (up to differences in upper and lower case).
2. Porter stemming [purple]: The two strings match after Porter stemming.
3. Edit distance [red]: The two strings have edit distance ≤ 2.

The four methods are: bot, V (VitaLITy), S_c (Semantic Scholar cited_for) and S_m (Semantic Scholar mentions). More details (code and outputs) for the bot method can be found in the GitHub.[14]

Comparisons within methods may be more meaningful than comparisons across methods. That is, the black V, purple V and red V in Figure 2 are all based on the same documents, but the three Vs are based on a different set of documents than the three bots. Comparisons over methods should be made with caution since not all methods are available for all papers, as shown in Figure 1.

As for comparisons within methods, it should not be surprising that precision and recall are higher for more generous matching criteria (red) than for stricter ones (black) across all 4 methods.

In general, the precision and recall are surprisingly low in Figure 2. We hope the community can propose methods that beat these baselines. Based on anecdotal experiments with citing sentences, we hoped that they could be used to improve precision and recall. Hopefully, the community will find a way to make a convincing case, though it is possible that citing sentences are more useful for finding short-hand that matters to contemporary audiences than for finding keywords that matter to authors of the original work.

[14]The bot method uses gpt-4o-mini with the prompt in subsection 3.3.

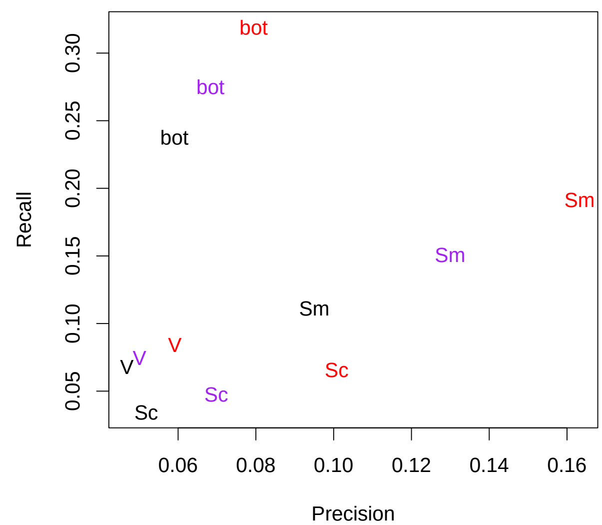

Figure 2: Results for 4 baselines, bot, V, S_c, S_m, and 3 types of matches: exact [black], Porter [purple], edit [red].

4 Conclusion and Future Work

We present KeyBench, a resource for a keyword extraction task. We provide author-supplied keywords as a gold standard, citing sentences, and evaluation methodology. The data set is constructed from three different sources: VitaLITy, Semantic Scholar, and by parsing ArXiv papers. Data and code are available (see footnote 5 and footnote 6) to encourage the community to beat our baselines, and combine KeyBench with other resources such as footnote 2.

The task defined in this paper is attractive in many ways, though it may not be the best way to demonstrate the power of citing sentences. We defined the gold standard in terms of keywords from authors, though it is clear that there are significant differences between keywords from authors and the words that contemporary audiences use in citing sentences to refer to those contributions. Citing sentences use terms like *Turing Machine* and the *Halting Problem* though these terms do not appear in the primary literature. Turing never named anything after himself, and his term for the Halting Problem, *Entscheidungsproblem*, did not survive the test of time, though his contribution remains very much "a thing" in the contemporary literature. If you want to know why a seminal paper is important to contemporary audiences, it is probably better to look in the contemporary literature than in the primary literature. The authors may not have appreciated just how important their work would become. As part of future work, we plan to use citing sentences to identify seminal papers and investigate how KeyBench can be used for academic search use cases.

References

[1] Alfred V Aho, Brian W Kernighan, and Peter J Weinberger. 2023. *The AWK programming language*. Addison-Wesley Professional.

[2] Hongye An, Arpit Narechania, Emily Wall, and Kai Xu. 2024. vitaLITy 2: Reviewing Academic Literature Using Large Language Models. Presented at the NLVIZ Workshop, IEEE VIS 2024. arXiv:2408.13450 [cs.HC] https://arxiv.org/abs/2408.13450

[3] Ricardo Campos, Vítor Mangaravite, Arian Pasquali, Alípio Mário Jorge, Célia Nunes, and Adam Jatowt. 2018. Yake! collection-independent automatic keyword extractor. In *European conference on information retrieval*. Springer, 806–810.

[4] Cornelia Caragea, Florin Adrian Bulgarov, Andreea Godea, and Sujatha Das Gollapalli. 2014. Citation-Enhanced Keyphrase Extraction from Research Papers: A Supervised Approach. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Alessandro Moschitti, Bo Pang, and Walter Daelemans (Eds.). Association for Computational Linguistics, Doha, Qatar, 1435–1446. doi:10.3115/v1/D14-1150

[5] Jun Chen, Xiaoming Zhang, Yu Wu, Zhao Yan, and Zhoujun Li. 2018. Keyphrase Generation with Correlation Constraints. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 4057–4066. doi:10.18653/v1/D18-1439

[6] Corina Florescu and Cornelia Caragea. 2017. PositionRank: An Unsupervised Approach to Keyphrase Extraction from Scholarly Documents. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Regina Barzilay and Min-Yen Kan (Eds.). Association for Computational Linguistics, Vancouver, Canada, 1105–1115. doi:10.18653/v1/P17-1102

[7] Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't Stop Pretraining: Adapt Language Models to Domains and Tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 8342–8360. doi:10.18653/v1/2020.acl-main.740

[8] Jui-Ting Huang, Ashish Sharma, Shuying Sun, Li Xia, David Zhang, Philip Pronin, Janani Padmanabhan, Giuseppe Ottaviano, and Linjun Yang. 2020. Embedding-based retrieval in facebook search. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, online, 2553–2561. https://api.semanticscholar.org/CorpusID:219965935

[9] Anette Hulth. 2003. Improved Automatic Keyword Extraction Given More Linguistic Knowledge. In *Proceedings of the 2003 Conference on Empirical Methods in Natural Language Processing*. ACL, online, 216–223. https://aclanthology.org/W03-1028/

[10] Su Nam Kim, Olena Medelyan, Min-Yen Kan, and Timothy Baldwin. 2010. SemEval-2010 Task 5 : Automatic Keyphrase Extraction from Scientific Articles. In *Proceedings of the 5th International Workshop on Semantic Evaluation*, Katrin Erk and Carlo Strapparava (Eds.). Association for Computational Linguistics, Uppsala, Sweden, 21–26. https://aclanthology.org/S10-1004/

[11] Daniel King, Doug Downey, and Daniel S. Weld. 2020. High-Precision Extraction of Emerging Concepts from Scientific Literature. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (2020). https://api.semanticscholar.org/CorpusID:219636085

[12] Rodney Michael Kinney, Chloe Anastasiades, Russell Authur, Iz Beltagy, Jonathan Bragg, Alexandra Buraczynski, Isabel Cachola, Stefan Candra, Yoganand Chandrasekhar, Arman Cohan, Miles Crawford, Doug Downey, Jason Dunkelberger, Oren Etzioni, Rob Evans, Sergey Feldman, Joseph Gorney, David W. Graham, F.Q. Hu, Regan Huff, Daniel King, Sebastian Kohlmeier, Bailey Kuehl, Michael Langan, Daniel Lin, Haokun Liu, Kyle Lo, Jaron Lochner, Kelsey MacMillan, Tyler C. Murray, Christopher Newell, Smita Rao, Shaurya Rohatgi, Paul Sayre, Shannon Zejiang Shen, Amanpreet Singh, Luca Soldaini, Shivashankar Subramanian, A. Tanaka, Alex D Wade, Linda M. Wagner, Lucy Lu Wang, Christopher Wilhelm, Caroline Wu, Jiangjiang Yang, Angele Zamarron, Madeleine van Zuylen, and Daniel S. Weld. 2023. The Semantic Scholar Open Data Platform. *ArXiv* abs/2301.10140 (2023). https://api.semanticscholar.org/CorpusID:256194545

[13] Wendy Lehnert, Claire Cardie, and Ellen Riloff. 1990. Analyzing Research papers using citation sentences. In *Proceedings of the Annual Meeting of the Cognitive Science Society*. online, 511–518.

[14] Kyle Lo, Lucy Lu Wang, Mark Neumann, Rodney Kinney, and Daniel Weld. 2020. S2ORC: The Semantic Scholar Open Research Corpus. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 4969–4983. doi:10.18653/v1/2020.acl-main.447

[15] Mitchell P. Marcus, Beatrice Santorini, and Mary Ann Marcinkiewicz. 1993. Building a Large Annotated Corpus of English: The Penn Treebank. *Comput. Linguistics* 19, 2 (1993), 313–330.

[16] Rada Mihalcea and Paul Tarau. 2004. TextRank: Bringing Order into Text. In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, Dekang Lin and Dekai Wu (Eds.). Association for Computational Linguistics, Barcelona, Spain, 404–411. https://aclanthology.org/W04-3252/

[17] Eirini Papagiannopoulou and Grigorios Tsoumakas. 2020. A review of keyphrase extraction. *WIREs Data Mining and Knowledge Discovery* 10, 2 (2020), e1339. arXiv:https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/widm.1339 doi:10.1002/widm.1339

[18] Amir Pouran Ben Veyseh, Franck Dernoncourt, Walter Chang, and Thien Huu Nguyen. 2021. MadDog: A Web-based System for Acronym Identification and Disambiguation. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, Dimitra Gkatzia and Djamé Seddah (Eds.). Association for Computational Linguistics, Online, 160–167. doi:10.18653/v1/2021.eacl-demos.20

[19] Ellen F Prince. 1981. *Toward a taxonomy of given-new information*. Academic Press.

[20] Vahed Qazvinian, Dragomir R. Radev, and Arzucan Özgür. 2010. Citation Summarization Through Keyphrase Extraction. In *Proceedings of the 23rd International Conference on Computational Linguistics (Coling 2010)*, Chu-Ren Huang and Dan Jurafsky (Eds.). Coling 2010 Organizing Committee, Beijing, China, 895–903. https://aclanthology.org/C10-1101/

[21] Ronald L Rivest, Adi Shamir, and Leonard Adleman. 1978. A method for obtaining digital signatures and public-key cryptosystems. *Commun. ACM* 21, 2 (1978), 120–126.

[22] Stuart Rose, Dave Engel, Nick Cramer, and Wendy Cowley. 2010. Automatic keyword extraction from individual documents. *Text mining: applications and theory* (2010), 1–20.

[23] Alan Mathison Turing. 1936. On computable numbers, with an application to the Entscheidungsproblem. *J. of Math* 58, 345-363 (1936), 5.

[24] Peter D. Turney. 2000. Learning Algorithms for Keyphrase Extraction. *Information Retrieval* 2 (2000), 303–336.

[25] Peter Willett. 2006. The Porter stemming algorithm: then and now. *Program* 40, 3 (2006), 219–223.

[26] Ian H Witten, Gordon W Paynter, Eibe Frank, Carl Gutwin, and Craig G Nevill-Manning. 1999. KEA: Practical automatic keyphrase extraction. In *Proceedings of the fourth ACM conference on Digital libraries*. ACM, online, 254–255.